



Qualification Pack

Crop and Plant Supervisor

Electives: Dragon Fruit Supervisor / Lemon Grass Supervisor

QP Code: CUTM/AGR/Q0401

Version: 1.0

NSQF Level: 5

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Qualification Pack

Contents

CUTM/AGR/Q0401: Crop and Plant Supervisor	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Elective 1: Dragon Fruit Supervisor</i>	3
<i>Elective 2: Lemon Grass Supervisor</i>	3
<i>Qualification Pack (QP) Parameters</i>	4
CUTM/AGR/N0409: Implement Health and Safety standards at Work Place	6
DGT/VSQ/N0102: Employability Skills (60 Hours)	11
CUTM/AGR/N0401: Plan and Deploy planting materials	19
CUTM/AGR/N0402: Suggest optimal cultivation practices with integrated farming	26
CUTM/AGR/N0403: Suggest integrated nutrient management in Dragon fruit cultivation	32
CUTM/AGR/N0404: Suggest integrated disease and pest management in Dragon fruit cultivation ...	39
CUTM/AGR/N0405: Plan and Suggest land Preparation and Planting	45
CUTM/AGR/N0406: Suggest Nutrient and Irrigation Management	49
CUTM/AGR/N0407: Suggest Weed Control and Plant Protection	55
CUTM/AGR/N0408: Post - Harvest storage and Lemongrass oil extraction	59
Assessment Guidelines and Weightage	63
<i>Assessment Guidelines</i>	63
<i>Assessment Weightage</i>	64
Acronyms	67
Glossary	68



Qualification Pack

CUTM/AGR/Q0401: Crop and Plant Supervisor

Brief Job Description

Crop and Plant Supervisor is responsible for understanding the market demand and market price to plan and development of appropriate land location, preparation of land, plantation, water management, intercultural operations, pest and disease management, harvesting and processing to deliver to the customer for Lemon Grass and other integrated plants.

Personal Attributes

A Crop and Plant Supervisor must be observant, practical, and knowledgeable in horticultural practices. They should possess strong planning and problem-solving abilities, along with a scientific mindset for integrated nutrient, pest, and irrigation management. Physical stamina, leadership, communication skills, and a commitment to sustainable farming and quality post-harvest practices are essential for success in this role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/AGR/N0409: Implement Health and Safety standards at Work Place](#)
2. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Electives(*mandatory to select at least one*):

Elective 1: Dragon Fruit Supervisor

A Dragon Fruit Supervisor is responsible for overseeing the cultivation, care, and post-harvest handling of dragon fruit crops. The role includes planning land preparation, deploying planting materials, managing nutrient and irrigation systems, and implementing integrated pest and disease control measures. The supervisor ensures optimal growth conditions, monitors crop health, and guides labor for field activities. Additionally, they are involved in harvesting, grading, and coordinating post-harvest storage or marketing. The position requires practical knowledge of sustainable horticultural practices, record-keeping, and the ability to train or guide farmers and workers in modern dragon fruit farming techniques.

1. [CUTM/AGR/N0401: Plan and Deploy planting materials](#)
2. [CUTM/AGR/N0402: Suggest optimal cultivation practices with integrated farming](#)
3. [CUTM/AGR/N0403: Suggest integrated nutrient management in Dragon fruit cultivation](#)
4. [CUTM/AGR/N0404: Suggest integrated disease and pest management in Dragon fruit cultivation](#)

Qualification Pack

Elective 2: Lemon Grass Supervisor

A Lemongrass Supervisor is responsible for managing the cultivation, maintenance, and post-harvest processes of lemongrass crops. The role includes planning land preparation, suggesting suitable nutrient and irrigation management, implementing weed and pest control measures, and monitoring crop health. The supervisor oversees harvesting operations and guides the extraction and storage of lemongrass oil. They ensure efficient use of resources, maintain quality standards, and may assist in marketing value-added products. The position requires practical knowledge of aromatic plant cultivation, sustainable farming practices, and the ability to lead field operations effectively.

1. [CUTM/AGR/N0405: Plan and Suggest land Preparation and Planting](#)
2. [CUTM/AGR/N0406: Suggest Nutrient and Irrigation Management](#)
3. [CUTM/AGR/N0407: Suggest Weed Control and Plant Protection](#)
4. [CUTM/AGR/N0408: Post - Harvest storage and Lemongrass oil extraction](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Modern Farming
Occupation	Field Crop
Country	India
NSQF Level	5
Credits	28
Aligned to NCO/ISCO/ISIC Code	NSQF/NCrF
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (agriculture) with NA of experience OR Completed 3 year diploma after 10th (Agriculture) with 1 Year of experience relevant experience OR 12th grade Pass with 2 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant experience



Qualification Pack

Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	30/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-05-AG-01978-2024-V1-CUTM
NQR Version	1.0

Qualification Pack

CUTM/AGR/N0409: Implement Health and Safety standards at Work Place

Description

This NOS focuses on maintaining personal hygiene, workplace safety, and environmental responsibility. It outlines procedures for disease prevention, safe equipment handling, waste disposal, and emergency response. The standard ensures that individuals follow preventive measures and safety protocols to create a clean, hazard-free, and health-conscious work environment.

Scope

The scope covers the following :

- The scope covers adherence to sanitation and hygiene norms, safe use of tools and chemicals, proper waste management, first aid administration, and compliance with government and workplace advisories. It aims to prepare individuals to work safely under routine and emergency situations, reducing risks to themselves, coworkers, and the environment.

Elements and Performance Criteria

Maintain personal hygiene

To be competent, the user/individual on the job must be able to:

- PC1.** wash hands, legs and face with soap/alcohol based sanitizer at reasonable intervals
- PC2.** wash the worn clothes with soap and sundry before use next time
- PC3.** ensure the face is covered with mask or three layers of cloth-piece
- PC4.** follow the workplace sanitization norms including distancing from sick people

Maintain clean and safe workplace

To be competent, the user/individual on the job must be able to:

- PC5.** carry out basic safety checks before operation of all tools, implements, and machinery and report identified hazards to the supervisor
- PC6.** follow the instructions mentioned on the labels of chemicals/pesticides/fumigants etc to avoid hazards
- PC7.** sanitize equipment, tools and machinery before and after use
- PC8.** dispose waste safely and correctly in the designated area
- PC9.** work in a manner which minimizes environmental damage, ensuring all procedures and instructions for controlling risks are followed
- PC10.** follow government / workplace advisories in case of outbreak of any disease/disaster

Administer appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements
- PC12.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques

Qualification Pack

PC13. report details of first aid administered in accordance with workplace procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Importance of personal hygiene in preventing infection and illness.
- KU2.** Use and effectiveness of masks and sanitizers.
- KU3.** Workplace safety norms, including sanitization and distancing practices.
- KU4.** Basic safety checks and hazard identification protocols.
- KU5.** Safe handling and usage instructions of chemicals and pesticides.
- KU6.** Procedures for cleaning and sanitizing tools and machinery.
- KU7.** Waste disposal methods and designated disposal areas.
- KU8.** Environmental safety practices and risk mitigation.
- KU9.** Government and workplace advisories related to health emergencies.
- KU10.** First aid techniques and workplace protocols for injury reporting.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to follow hygiene and sanitization routines consistently.
- GS2.** Situational awareness to identify unsafe practices or potential hazards.
- GS3.** Communication skills to report hazards and emergencies accurately.
- GS4.** Decision-making during accidents or outbreaks.
- GS5.** Capability to use PPE and follow safety signs/instructions.
- GS6.** Time management in emergency response or first aid delivery.
- GS7.** Teamwork while maintaining workplace health and safety norms.
- GS8.** Basic technical skill to perform equipment safety checks.
- GS9.** Problem-solving in waste management and chemical handling.
- GS10.** Record-keeping for first aid and incident reporting.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal hygiene</i>	10	10	-	9
PC1. wash hands, legs and face with soap/alcohol based sanitizer at reasonable intervals	2	4	-	2
PC2. wash the worn clothes with soap and sundry before use next time	2	2	-	2
PC3. ensure the face is covered with mask or three layers of cloth-piece	2	2	-	3
PC4. follow the workplace sanitization norms including distancing from sick people	4	2	-	2
<i>Maintain clean and safe workplace</i>	10	20	-	15
PC5. carry out basic safety checks before operation of all tools, implements, and machinery and report identified hazards to the supervisor	2	2	-	3
PC6. follow the instructions mentioned on the labels of chemicals/pesticides/fumigants etc to avoid hazards	2	2	-	2
PC7. sanitize equipment, tools and machinery before and after use	2	2	-	3
PC8. dispose waste safely and correctly in the designated area	2	4	-	3
PC9. work in a manner which minimizes environmental damage, ensuring all procedures and instructions for controlling risks are followed	1	5	-	2
PC10. follow government / workplace advisories in case of outbreak of any disease/disaster	1	5	-	2
<i>Administer appropriate emergency procedures</i>	10	10	-	6
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements	2	6	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques	2	2	-	2
PC13. report details of first aid administered in accordance with workplace procedures	6	2	-	2
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0409
NOS Name	Implement Health and Safety standards at Work Place
Sector	Agriculture
Sub-Sector	
Occupation	Field Crop
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

Qualification Pack

CUTM/AGR/N0401: Plan and Deploy planting materials

Description

This NOS involves selecting appropriate land, testing and preparing soil, laying out fields using standard systems (square, rectangle, contour), and deploying required tools, equipment, and support structures to ensure optimal dragon fruit cultivation.

Scope

The scope covers the following :

- The scope of the NOS includes preparing suitable land, testing soil for nutrient levels, selecting the appropriate crop variety, planning field layout, organizing required tools and equipment, and deploying planting materials effectively to ensure optimal growth and productivity of crops like dragon fruit.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Leveling of the land appropriately for cultivation
- PC2.** Creation of proper drainage systems
- PC3.** Selection of the appropriate field layout (square system, rectangle system, triangle system, quincunxial system, contour system) based on the variety, yield expected and canopy arrangement
- PC4.** Placement of the markers at the site at appropriate distances
- PC5.** Creating the pits at appropriate depths
- PC6.** Irrigating the pits at right amounts and allow dryness
- PC7.** Identification of the Agro-climatic zone in which cultivation is taken up
- PC8.** Enlisting the varieties that are suitable to the zone
- PC9.** Selection of economically attractive variety
- PC10.** Identification of various vendors / suppliers (including government nurseries / Universities) of the planting material, if the farmer is not preparing the planting material
- PC11.** Assuring the quality of planting material from each source in terms of 'free from pests and diseases', 'survival rate' etc.
- PC12.** Determining the prevailing market rates for the planting material
- PC13.** Procuring the planting material
- PC14.** Identification of an appropriate storage space (free of infestation and having congenial climatic conditions for the planting material)
- PC15.** Storage of the planting material (if there is time lag between procurement and planting)
- PC16.** Preparation of the planting material appropriately through practices suitable to variety
- PC17.** Usage of vegetative propagation to get true to type planting material
- PC18.** Treatment of the planting material appropriately
- PC19.** Storage of planting material hygienically

Qualification Pack

- PC20.** Ascertaining the right storage time for planting material based on the practice adopted
- PC21.** Ascertaining the prevalent pests and diseases of Dragon fruit in a given agro- climatic zone
- PC22.** Ascertaining the appropriate pesticides and fungicide available for planting material treatment
- PC23.** Treatment of the planting material as per the dosage recommended by the State Agriculture University/Department or as prescribed by the pesticide manufacturer
- PC24.** Reading of the safety precautions in the label/instruction that comes with the pesticide
- PC25.** Keeping ready all the necessary first aids as suggested in the safety measures
- PC26.** Usage all the necessary safety material and follow all the preventive measures
- PC27.** Avoid any injury during use / application of pesticide
- PC28.** Usage of appropriate irrigation method and follow irrigation schedule based on the availability of the resource, climate, age of plant and soil type etc.
- PC29.** Adoption of methods of precision farming (drip/basin irrigation)
- PC30.** Usage of existing resource efficiently say fertigation etc
- PC31.** Management of drought and practicing soil moisture conservation by mulching check that the site has easy access to the local market
- PC32.** Selection of crops that can be taken as inter/mixed crop based on the climatic requirement of the inter/mixed crop, irrigation facilities and soil type, market and price
- PC33.** Consideration of the canopy size, age and spacing of the Dragon fruit at the time of selection of inter/mixed crop
- PC34.** Consideration of green crops and cover crops that may also be raised to increase the organic matter content of the soil and to prevent soil erosion
- PC35.** Consideration in taking multiple crops depending upon the suitability.
- PC36.** Application of recommended quantity of water and manures and fertilizers to the intercrops separately.
- PC37.** Pruning as per requirement and procedures
- PC38.** Removal of dead plant parts
- PC39.** Tillage operation
- PC40.** Utilization of plant waste
- PC41.** Application of organic manure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** how to identify suitable land based on climate, soil type, and topography essential for cultivating crops like dragon fruit.
- KU2.** how to test and interpret soil pH and nutrient levels using kits or lab analysis to determine land readiness.
- KU3.** the methods for clearing, levelling, and improving drainage of land, including the use of raised beds or mounds.
- KU4.** various planting layout systems such as square, rectangular, triangular, and contour systems, and their application in efficient land use.

Qualification Pack

- KU5.** how to calculate plant and row spacing (e.g., 8 feet apart) and align rows for maximum sunlight exposure.
- KU6.** the familiarity with erecting trellises or posts for plant support, especially for crops like dragon fruit that require vertical growth structures.
- KU7.** the identification and operation key tools such as shovels, spades, mulching machines, pruning tools, irrigation systems, and soil testers for effective plantation management.
- KU8.** the organization of farm inputs like pest control kits, fertilizers, protective gear, and first-aid kits for safe and productive operations.
- KU9.** the importance of proper waste disposal, eco-friendly practices, and soil conservation techniques during land deployment.
- KU10.** the basic knowledge of materials and processes used for storing and packaging agricultural produce post-planting.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** plan and implement appropriate field layouts such as square, rectangular, triangular, or contour systems for effective plantation.
- GS2.** do the testing soil for pH, fertility, and nutrient content, followed by clearing, levelling, and preparing land for optimal water drainage and root development.
- GS3.** do the deployment of spacing markers and support systems like trellises to enhance plant growth and sun exposure, particularly for dragon fruit cultivation.
- GS4.** competence in using various agricultural tools such as shovels, spades, soil testing kits, irrigation tools, and planting layout instruments.
- GS5.** select suitable plant varieties based on agro-climatic conditions, soil properties, and economic viability for high yield.
- GS6.** utilize the skill in organizing materials, manpower, and logistics for timely deployment of planting materials, ensuring synchronization with planting seasons.
- GS7.** choose locations based on climate, drainage, sunlight orientation (north-south rows), and potential natural threats.
- GS8.** do best practices and technical standards for sustainable planting, including pest management readiness, proper irrigation alignment, and nutrient support.
- GS9.** troubleshoot land issues like waterlogging or stony soil and adapt layouts or materials accordingly to ensure planting success.
- GS10.** instruct field workers, coordinate with suppliers, and maintain proper documentation and reporting related to land and planting material deployment.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. Leveling of the land appropriately for cultivation	2	2	-	1
PC2. Creation of proper drainage systems	2	2	-	1
PC3. Selection of the appropriate field layout (square system, rectangle system, triangle system, quincunxial system, contour system) based on the variety, yield expected and canopy arrangement	2	2	-	1
PC4. Placement of the markers at the site at appropriate distances	2	2	-	1
PC5. Creating the pits at appropriate depths	2	2	-	1
PC6. Irrigating the pits at right amounts and allow dryness	1	3	-	1
PC7. Identification of the Agro-climatic zone in which cultivation is taken up	1	-	-	1
PC8. Enlisting the varieties that are suitable to the zone	1	-	-	1
PC9. Selection of economically attractive variety	1	1	-	-
PC10. Identification of various vendors / suppliers (including government nurseries / Universities) of the planting material, if the farmer is not preparing the planting material	1	2	-	-
PC11. Assuring the quality of planting material from each source in terms of 'free from pests and diseases', 'survival rate' etc.	1	2	-	1
PC12. Determining the prevailing market rates for the planting material	1	2	-	1
PC13. Procuring the planting material	1	2	-	1
PC14. Identification of an appropriate storage space (free of infestation and having congenial climatic conditions for the planting material)	-	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Storage of the planting material (if there is time lag between procurement and planting)	1	2	-	1
PC16. Preparation of the planting material appropriately through practices suitable to variety	1	2	-	-
PC17. Usage of vegetative propagation to get true to type planting material	1	-	-	-
PC18. Treatment of the planting material appropriately	-	2	-	-
PC19. Storage of planting material hygienically	1	2	-	1
PC20. Ascertaining the right storage time for planting material based on the practice adopted	1	-	-	1
PC21. Ascertaining the prevalent pests and diseases of Dragon fruit in a given agro- climatic zone	1	-	-	1
PC22. Ascertaining the appropriate pesticides and fungicide available for planting material treatment	1	-	-	1
PC23. Treatment of the planting material as per the dosage recommended by the State Agriculture University/Department or as prescribed by the pesticide manufacturer	-	2	-	1
PC24. Reading of the safety precautions in the label/instruction that comes with the pesticide	-	-	-	1
PC25. Keeping ready all the necessary first aids as suggested in the safety measures	-	-	-	1
PC26. Usage all the necessary safety material and follow all the preventive measures	-	-	-	-
PC27. Avoid any injury during use / application of pesticide	1	-	-	-
PC28. Usage of appropriate irrigation method and follow irrigation schedule based on the availability of the resource, climate, age of plant and soil type etc.	-	2	-	1
PC29. Adoption of methods of precision farming (drip/basin irrigation)	-	1	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC30. Usage of existing resource efficiently say fertigation etc	-	-	-	1
PC31. Management of drought and practicing soil moisture conservation by mulching check that the site has easy access to the local market	-	-	-	1
PC32. Selection of crops that can be taken as inter/mixed crop based on the climatic requirement of the inter/mixed crop, irrigation facilities and soil type, market and price	1	-	-	1
PC33. Consideration of the canopy size, age and spacing of the Dragon fruit at the time of selection of inter/mixed crop	-	-	-	1
PC34. Consideration of green crops and cover crops that may also be raised to increase the organic matter content of the soil and to prevent soil erosion	-	2	-	1
PC35. Consideration in taking multiple crops depending upon the suitability.	1	-	-	1
PC36. Application of recommended quantity of water and manures and fertilizers to the intercrops separately.	-	-	-	-
PC37. Pruning as per requirement and procedures	-	-	-	-
PC38. Removal of dead plant parts	1	-	-	1
PC39. Tillage operation	-	-	-	-
PC40. Utilization of plant waste	1	-	-	1
PC41. Application of organic manure	-	1	-	-
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0401
NOS Name	Plan and Deploy planting materials
Sector	Agriculture
Sub-Sector	
Occupation	Field Crop
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQF Clearance Date	31/01/2024

Qualification Pack

CUTM/AGR/N0402: Suggest optimal cultivation practices with integrated farming

Description

This NOS equips candidates to plan and implement optimal dragon fruit cultivation practices within an integrated farming system. It promotes diversification and sustainability by combining fruit crops with compatible agricultural or allied activities.

Scope

The scope covers the following :

- Involves the assessment, planning, and execution of integrated farming practices around dragon fruit cultivation. It enhances productivity, minimizes risks, and promotes income diversification using sustainable farming models tailored to local conditions.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Soil sample is taken as per the recommended sampling procedure from the field
- PC2.** The soil is dried and samples are prepared as per procedure Ascertain the prices of various inputs in the market
- PC3.** Soil sample is packed, labelled and submitted to nearby soil testing laboratory for analysis
- PC4.** The soil analysis report is collected from the lab
- PC5.** Recommended organic and inorganic fertilizer dosage is collected from agriculture department based on the soil analysis report
- PC6.** To achieve higher productivity, regular manuring from the first year of planting is essential
- PC7.** Application of the manure to the soil as per recommended dosage and procedure
- PC8.** Selection of suitable green manure/ eco-friendly botanicals/bio-fertilizer that can be applied
- PC9.** Adoption of various cultural practices that enhances the soil nutrient status for the benefit of Dragon Fruit crop stand
- PC10.** Fertilizer application for macro and micro nutrients as per the recommended dosage, timing, placement locations and method of application
- PC11.** Practicing weed control as per recommendation
- PC12.** Removal of rogue infected plant/ plant parts
- PC13.** Performing weeding manually / application of herbicides as per the recommendations and instructions
- PC14.** Identification of the different types of pests associated in Dragon Fruit cultivation
- PC15.** Identification of the stage of crop and pest incidence – pest calendar
- PC16.** Identification of the signs and symptoms of damage
- PC17.** Identification of the pest life cycle – estimation of the duration
- PC18.** Identification of the natural enemies of the pests
- PC19.** Identification of the different types of diseases associated in Dragon Fruit cultivation

Qualification Pack

- PC20.** Identification of specific problems in Dragon Fruit plant
- PC21.** Identification of physiological disorders (lack of major nutrients/minor nutrients)
- PC22.** Identification of the crop stage and disease incidence – disease calendar
- PC23.** Identification of the signs and symptoms of different diseases
- PC24.** Identification of the mode of transmission of disease
- PC25.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements
- PC26.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques
- PC27.** report details of first aid administered in accordance with workplace procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the appropriate varieties of dragon fruit suitable for different agro-climatic zones and their economic viability.
- KU2.** the gain knowledge on the procurement, treatment, and storage of planting materials, including grafting, budding, and vegetative propagation methods.
- KU3.** how to prepare and test soil samples, interpret lab results, and adjust soil conditions using nutrients, lime, or gypsum.
- KU4.** the application of organic and inorganic fertilizers, composting, and using mulching techniques to improve soil fertility.
- KU5.** the knowledge of drip and basin irrigation systems, fertigation techniques, and efficient water-use practices tailored to plant and soil needs.
- KU6.** how to plan and implement intercropping and mixed cropping systems to enhance biodiversity, improve yield, and reduce pest outbreaks.
- KU7.** the awareness of pest and disease cycles, their natural predators, and the importance of using resistant varieties and biological control methods.
- KU8.** sustainable and organic farming methods, such as using cover crops, green manure, and minimizing chemical inputs.
- KU9.** about plant spacing, row alignment, and the use of support structures or trellises for optimal plant growth and sunlight exposure.
- KU10.** the safe use of pesticides/fungicides, importance of personal protective equipment (PPE), and first aid in farm operations.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Assess soil, climate, and water availability to recommend suitable cultivation methods for dragon fruit and compatible intercrops.
- GS2.** Plan and supervise integrated farming systems involving crops, livestock, or fishery alongside dragon fruit cultivation for resource optimization.

Qualification Pack

- GS3.** Recommend appropriate planting techniques including spacing, trellising, and pit preparation for healthy plant growth.
- GS4.** apply organic farming methods, including the use of compost, green manure, and biofertilizers to improve soil health and yield.
- GS5.** be proficient of selecting and operating irrigation systems such as drip and basin irrigation, ensuring optimal water use efficiency.
- GS6.** Monitor field operations and suggest improvements based on growth stage, season, and resource availability.
- GS7.** Coordinate water and nutrient use across systems.
- GS8.** prepare, label, and analyse soil samples to create precise nutrient management plans.
- GS9.** use agricultural tools such as mulchers, weed trimmers, soil testing kits, and protective gear for daily operations.
- GS10.** obtain the commitment to eco-friendly cultivation by adopting practices that conserve soil moisture and reduce erosion.
- GS11.** maintain proper records of planting schedules, nutrient inputs, pest/disease outbreaks, and ensure adherence to agricultural norms and guidelines.
- GS12.** Solve practical challenges in integrated setups.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Soil sample is taken as per the recommended sampling procedure from the field	2	3	-	-
PC2. The soil is dried and samples are prepared as per procedure Ascertain the prices of various inputs in the market	2	3	-	-
PC3. Soil sample is packed, labelled and submitted to nearby soil testing laboratory for analysis	2	3	-	-
PC4. The soil analysis report is collected from the lab	2	3	-	-
PC5. Recommended organic and inorganic fertilizer dosage is collected from agriculture department based on the soil analysis report	2	3	-	-
PC6. To achieve higher productivity, regular manuring from the first year of planting is essential	2	3	-	-
PC7. Application of the manure to the soil as per recommended dosage and procedure	-	-	-	-
PC8. Selection of suitable green manure/ eco-friendly botanicals/bio-fertilizer that can be applied	2	3	-	-
PC9. Adoption of various cultural practices that enhances the soil nutrient status for the benefit of Dragon Fruit crop stand	2	3	-	-
PC10. Fertilizer application for macro and micro nutrients as per the recommended dosage, timing, placement locations and method of application	1	4	-	-
PC11. Practicing weed control as per recommendation	1	4	-	-
PC12. Removal of rogue infected plant/ plant parts	1	4	-	-
PC13. Performing weeding manually / application of herbicides as per the recommendations and instructions	3	7	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Identification of the different types of pests associated in Dragon Fruit cultivation	2	-	-	-
PC15. Identification of the stage of crop and pest incidence – pest calendar	2	3	-	-
PC16. Identification of the signs and symptoms of damage	2	-	-	-
PC17. Identification of the pest life cycle – estimation of the duration	2	3	-	-
PC18. Identification of the natural enemies of the pests	2	3	-	-
PC19. Identification of the different types of diseases associated in Dragon Fruit cultivation	2	3	-	-
PC20. Identification of specific problems in Dragon Fruit plant	-	-	-	-
PC21. Identification of physiological disorders (lack of major nutrients/minor nutrients)	2	3	-	-
PC22. Identification of the crop stage and disease incidence – disease calendar	2	-	-	-
PC23. Identification of the signs and symptoms of different diseases	1	-	-	-
PC24. Identification of the mode of transmission of disease	1	2	-	-
PC25. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements	-	-	-	-
PC26. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques	-	-	-	-
PC27. report details of first aid administered in accordance with workplace procedures	-	-	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0402
NOS Name	Suggest optimal cultivation practices with integrated farming
Sector	Agriculture
Sub-Sector	
Occupation	Field Crop
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/AGR/N0403: Suggest integrated nutrient management in Dragon fruit cultivation

Description

This NOS focuses on designing and implementing an integrated nutrient management (INM) strategy for dragon fruit cultivation. It combines organic, inorganic, and biological sources to enhance crop health, yield, and sustainability.

Scope

The scope covers the following :

- Includes soil testing, nutrient planning, scheduling, application supervision, and monitoring outcomes. It emphasizes balanced and sustainable fertilization practices that improve soil fertility and support long-term crop productivity in dragon fruit farming.

Elements and Performance Criteria

Managing pre harvesting of Dragon Fruit

To be competent, the user/individual on the job must be able to:

- PC1.** Calculating the number of days from the flowering to have an estimate on the ripening day
- PC2.** Ascertaining the right indicators when the Dragon Fruit is ready for harvesting by checking the colour, the flattening in the fruit, roundness, etc
- PC3.** Encouragement for natural ripening

Managing Dragon Fruit harvesting, post harvesting and storage

To be competent, the user/individual on the job must be able to:

- PC4.** Harvesting the crop at the right time
- PC5.** Usage of fruit harvesters wherever applicable
- PC6.** Keep the harvested Dragon Fruits away from heat and moisture
- PC7.** Transport the harvested Dragon Fruits to the storage site
- PC8.**
 - Use wooden or cardboard boxes / plastic crates / bamboo baskets / gunny bags or any other material as per the market
 - requirement and demand and for handling in an efficient and effective manner
- PC9.** Choose means of transportation, say by trucks/lorries / rail wagons for distant markets based on the requirement and cost dynamics involved.
- PC10.** Decision on storage or sale based on the prevailing market rates across the markets and the cost of storage

Marketing of the produce

To be competent, the user/individual on the job must be able to:

- PC11.** Identification of the right market for sale of produce
- PC12.** Analyzing the right time for sale considering the periodical demand for the product
- PC13.** Coordination and negotiation with procurement assistant of the buyer for best price

Making Post harvest products from Dragon Fruit

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC14.** Setting up of small scale post-harvest unit
- PC15.** Jam, Ready to Serve (RTS) and Squash from dragon fruit pulp
- PC16.** Proper storage of the products
- PC17.** Finding out proper market channel (Malls/ e-commerce)
- PC18.** Financial analysis out of post-harvest products of Dragon fruits.
- PC19.** Ascertain total cost of production (land, production practices, labor, equipment , fuel, administrative cost etc.)
- PC20.** Maintain records of investment and expenditure
- PC21.** Maintain necessary books of accounts
- PC22.** Identify government schemes and their eligibility for availing themselves of the same
- PC23.** Identification of the nearest market
- PC24.** Identification of local traders, mandis in the villages and nearby and compare the rates
- PC25.** Identification of market rates of the produce season wise
- PC26.** Arrangement of cost-effective transportation of produce to the market

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Nutrient requirements and deficiency symptoms in dragon fruit.
- KU2.** Soil testing and analysis techniques.
- KU3.** Organic, inorganic, and bio-fertilizer options.
- KU4.** Schedule and dosage of nutrient application.
- KU5.** Drip fertigation and foliar feeding methods.
- KU6.** Nutrient use efficiency principles.
- KU7.** Role of compost, vermicompost, and green manure.
- KU8.** Impact of nutrient imbalance on fruit quality.
- KU9.** Record-keeping and nutrient budgeting.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Recommend fertilizer types and dosages based on soil tests.
- GS2.** Plan and supervise timely nutrient application.
- GS3.** Train farmers on composting and organic inputs.
- GS4.** Ensure proper calibration of fertigation units.
- GS5.** Monitor plant response and adjust nutrient schedule.
- GS6.** Maintain nutrient application logs.
- GS7.** Reduce dependency on chemical inputs.
- GS8.** Coordinate with agri-input dealers and extension agents.



Qualification Pack

- GS9.** Evaluate cost-effectiveness of nutrient plans.
- GS10.** Promote nutrient recycling from farm waste.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Managing pre harvesting of Dragon Fruit</i>	6	9	-	-
PC1. Calculating the number of days from the flowering to have an estimate on the ripening day	2	3	-	-
PC2. Ascertaining the right indicators when the Dragon Fruit is ready for harvesting by checking the colour, the flattening in the fruit, roundness, etc	2	3	-	-
PC3. Encouragement for natural ripening	2	3	-	-
<i>Managing Dragon Fruit harvesting, post harvesting and storage</i>	13	20	-	-
PC4. Harvesting the crop at the right time	2	3	-	-
PC5. Usage of fruit harvesters wherever applicable	2	3	-	-
PC6. Keep the harvested Dragon Fruits away from heat and moisture	2	3	-	-
PC7. Transport the harvested Dragon Fruits to the storage site	2	3	-	-
PC8. • Use wooden or cardboard boxes / plastic crates / bamboo baskets / gunny bags or any other material as per the market • requirement and demand and for handling in an efficient and effective manner	2	3	-	-
PC9. Choose means of transportation, say by trucks/lorries / rail wagons for distant markets based on the requirement and cost dynamics involved.	1	2	-	-
PC10. Decision on storage or sale based on the prevailing market rates across the markets and the cost of storage	2	3	-	-
<i>Marketing of the produce</i>	3	7	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Identification of the right market for sale of produce	1	3	-	-
PC12. Analyzing the right time for sale considering the periodical demand for the product	1	3	-	-
PC13. Coordination and negotiation with procurement assistant of the buyer for best price	1	1	-	-
<i>Making Post harvest products from Dragon Fruit</i>	18	24	-	-
PC14. Setting up of small scale post-harvest unit	1	1	-	-
PC15. Jam, Ready to Serve (RTS) and Squash from dragon fruit pulp	1	1	-	-
PC16. Proper storage of the products	1	1	-	-
PC17. Finding out proper market channel (Malls/ e-commerce)	1	1	-	-
PC18. Financial analysis out of post-harvest products of Dragon fruits.	1	2	-	-
PC19. Ascertain total cost of production (land, production practices, labor, equipment , fuel, administrative cost etc.)	1	2	-	-
PC20. Maintain records of investment and expenditure	1	3	-	-
PC21. Maintain necessary books of accounts	2	2	-	-
PC22. Identify government schemes and their eligibility for availing themselves of the same	1	3	-	-
PC23. Identification of the nearest market	2	2	-	-
PC24. Identification of local traders, mandis in the villages and nearby and compare the rates	2	2	-	-
PC25. Identification of market rates of the produce season wise	2	2	-	-
PC26. Arrangement of cost-effective transportation of produce to the market	2	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0403
NOS Name	Suggest integrated nutrient management in Dragon fruit cultivation
Sector	Agriculture
Sub-Sector	
Occupation	Field Crop
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/AGR/N0404: Suggest integrated disease and pest management in Dragon fruit cultivation

Description

This NOS equips learners to identify pests and diseases, understand their life cycles, and implement eco-friendly control strategies like pruning, sanitation, and using beneficial insects, promoting sustainable farming and crop protection in Dragon fruit cultivation.

Scope

The scope covers the following :

- The scope of the NOS includes identifying pest and disease types, analyzing crop symptoms, understanding pest life cycles, applying eco-friendly control measures, and promoting sustainable, preventive practices using resistant varieties, pruning, and beneficial insects for healthier yields.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.**
 - Understanding of the different sources of information at village-level through other farmers, neighbors, relatives,
 - agricultural extension workers, agriculture specialists, concerned government and private departments like gram
 - panchayat, co-operative societies and SHGs etc.
- PC2.** Identification of different sources of information at market level through commission agents, mandi, samitis and input dealers
- PC3.** Identification of different sources of information through media sources like radio, newspapers, television, magazine internet, SMS in mobile phones etc
- PC4.** Identify the appropriate sources of specific market information and proper ways to collect the required information
- PC5.** Identify the reliable source of information
- PC6.** Information gathering regarding subsidy provided by state government of NHB (National Horticulture Board)
- PC7.** Information about bank loans for capitals
- PC8.** Any financial assistance provided by Govt. of Odisha and Central Government.
- PC9.**
 - Ascertain methods of collecting information through personal visit, telephone, internet and published reports,
 - magazines and articles, workshops, attending seminars and training by agriculture extension service providers
- PC10.** Ascertain periodicity and cost of assessing market information
- PC11.** Ascertain availability and non-availability of specific market information
- PC12.** Perform documentation for analyzing market information
- PC13.** Evaluate the authenticity of information received
- PC14.** Analyze the information for taking decision

Qualification Pack

- PC15.** Utilize market information for taking cost effective production decisions
- PC16.** Understand quality-wise and variety-wise prices of different products such as planting material, agro-chemicals, fertilizer, etc
- PC17.** Use market information and decide on crop and area to be sown which could result in better productivity for the season
- PC18.**
- Utilize market information for taking effective pre-harvesting decisions like preparation of planting material, land
 - preparation, nutrition management, weed management, pest and diseases management and irrigation management
- PC19.** Utilize market information for appropriate post-harvesting decision like drying, grading, bagging, transportation, processing and storage.
- PC20.** Decision making on marketing parameters like where to sell, when to sell, to whom to sell and what quantity to sell etc. which leads to profit
- PC21.** Understanding the benefits derived from market information
- PC22.** Making of projections/future price movements through information
- PC23.** Understanding the price fluctuations in markets and take appropriate decision

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the identification of various pests affecting dragon fruit, their life stages, signs of damage, and seasonal occurrence using a pest calendar.
- KU2.** about beneficial insects (e.g., ladybugs, parasitoid wasps) that can help in controlling pest populations naturally.
- KU3.** to identify different diseases and physiological disorders caused by nutrient deficiencies (major and minor) and their symptoms.
- KU4.** the disease lifecycle, modes of transmission, and the crop stages at which different diseases are most likely to occur.
- KU5.** the importance of hygiene in cultivation areas, pruning techniques to remove infected parts, and disinfecting tools to prevent spread.
- KU6.** about disease-resistant dragon fruit varieties that help in reducing dependency on chemicals.
- KU7.** about the recommended chemical pesticides and their safe and appropriate usage for managing diseases and pests.
- KU8.** the use of soil conditioners like gypsum or sulphur to manage soil-related disorders affecting crop health.
- KU9.** the concept of IPM involving a combination of cultural, biological, and chemical practices for sustainable pest and disease control.
- KU10.** the safety precautions in handling, storing, and applying chemicals, including the importance of PPE and first aid awareness.

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** identify common pests and diseases affecting dragon fruit through signs, symptoms, pest life cycles, and disease calendars.
- GS2.** apply diagnostic skills to recognize physiological disorders related to nutrient deficiencies and environmental stress.
- GS3.** understand and utilize beneficial insects (e.g., ladybugs, parasitoid wasps) to naturally control pest populations.
- GS4.** demonstrate correct usage of chemical pesticides and fungicides following safety norms and manufacturer guidelines.
- GS5.** execute cultural practices like pruning of affected parts and maintaining cleanliness to prevent the spread of infections.
- GS6.** be proficient in using and disinfecting pruning tools to avoid disease transmission between plants.
- GS7.** implement IPM strategies combining mechanical, biological, and chemical methods for sustainable crop protection.
- GS8.** analyze pest calendars and field observations to plan timely interventions and preventive measures.
- GS9.** be capable of guiding farm laborers on pest control protocols and coordinating with agronomy experts for decision-making.
- GS10.** maintain records of pest/disease incidence and control measures, ensure alignment with regulatory standards for crop protection.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. - Understanding of the different sources of information at village-level through other farmers, neighbors, relatives, - agricultural extension workers, agriculture specialists, concerned government and private departments like gram - panchayat, co-operative societies and SHGs etc.	2	3	-	-
PC2. Identification of different sources of information at market level through commission agents, mandi, samitis and input dealers	2	3	-	-
PC3. Identification of different sources of information through media sources like radio, newspapers, television, magazine internet, SMS in mobile phones etc	2	3	-	-
PC4. Identify the appropriate sources of specific market information and proper ways to collect the required information	2	3	-	-
PC5. Identify the reliable source of information	2	3	-	-
PC6. Information gathering regarding subsidy provided by state government of NHB (National Horticulture Board)	2	3	-	-
PC7. Information about bank loans for capitals	2	3	-	-
PC8. Any financial assistance provided by Govt. of Odisha and Central Government.	2	3	-	-
PC9. - Ascertain methods of collecting information through personal visit, telephone, internet and published reports, - magazines and articles, workshops, attending seminars and training by agriculture extension service providers	2	3	-	-
PC10. Ascertain periodicity and cost of assessing market information	-	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Ascertain availability and non-availability of specific market information	-	3	-	-
PC12. Perform documentation for analyzing market information	2	3	-	-
PC13. Evaluate the authenticity of information received	1	3	-	-
PC14. Analyze the information for taking decision	1	3	-	-
PC15. Utilize market information for taking cost effective production decisions	1	3	-	-
PC16. Understand quality-wise and variety-wise prices of different products such as planting material, agro-chemicals, fertilizer, etc	1	3	-	-
PC17. Use market information and decide on crop and area to be sown which could result in better productivity for the season	1	3	-	-
PC18. - Utilize market information for taking effective pre-harvesting decisions like preparation of planting material, land - preparation, nutrition management, weed management, pest and diseases management and irrigation management	1	4	-	-
PC19. Utilize market information for appropriate post-harvesting decision like drying, grading, bagging, transportation, processing and storage.	-	3	-	-
PC20. Decision making on marketing parameters like where to sell, when to sell, to whom to sell and what quantity to sell etc. which leads to profit	1	3	-	-
PC21. Understanding the benefits derived from market information	1	3	-	-
PC22. Making of projections/future price movements through information	1	3	-	-
PC23. Understanding the price fluctuations in markets and take appropriate decision	1	3	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0404
NOS Name	Suggest integrated disease and pest management in Dragon fruit cultivation
Sector	Agriculture
Sub-Sector	
Occupation	Field Crop
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/AGR/N0405: Plan and Suggest land Preparation and Planting

Description

This NOS about the assess site suitability for Lemongrass cultivation, ensure irrigation access, and perform primary tilling. It includes preparing soil, creating ridges and furrows, adjusting pH, and installing irrigation systems for effective planting.

Scope

The scope covers the following :

- This NOS focuses on preparing suitable land for cultivating medicinal and aromatic plants like lemongrass. It includes site selection, soil assessment, tillage, field layout, irrigation setup, planting time, and slip planting techniques, ensuring effective crop establishment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Interact with the Customer on their requirement and decide different types of plants for integrated farming
- PC2.** Ensure proper sunlight exposure for at least 6-8 hours a day.
- PC3.** Clear the land of debris and weeds.
- PC4.** Plough or till the soil to a depth of 8-10 inches.
- PC5.** Incorporate well-rotted organic matter to improve soil fertility.
- PC6.** Use healthy and disease free lemon grass saplings.
- PC7.** Plant at a spacing of 2-3 feet between rows and 1-2 feet between plants.
- PC8.** Plant during the rainy season or provide adequate irrigation.
- PC9.** Ensures proper soil conditioning and preparation for optimal plant growth

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of selecting a cultivation site with favourable climatic conditions, soil health, and accessibility to water, labour, and inputs for optimal growth of medicinal and aromatic plants like lemongrass.
- KU2.** the process of checking soil pH and making amendments using lime or gypsum to create suitable conditions for planting.
- KU3.** the importance of planning and installing efficient irrigation or fertigation systems to support healthy crop growth and water use efficiency.
- KU4.** how to be familiar with primary and secondary tilling techniques using appropriate farm machinery to prepare the land for planting.
- KU5.** the necessity of creating drainage channels, ridges, and furrows to prevent waterlogging and ensure uniform water flow.

Qualification Pack

- KU6.** the ideal time for planting slips based on moisture content, precipitation, and humidity, as well as the spacing and layout required for effective planting.
- KU7.** how to procure and prepare planting material (slips), including varietal selection and treatment before sowing.
- KU8.** how to be aware of potential risks and hazards associated with site conditions and implement appropriate preventive measures for successful cultivation.
- KU9.** the use of farm tools and equipment required for levelling, tilling, and planting operations effectively.
- KU10.** how to design field layouts according to terrain and planting systems for optimal plant growth and resource utilization.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** assess land for lemongrass and medicinal plant cultivation by evaluating climatic conditions, soil quality, water availability, and accessibility.
- GS2.** check soil pH and recommend amendments such as lime or gypsum to optimize soil conditions for planting.
- GS3.** be proficient in leveling land, conducting primary and secondary tilling, and creating ridges, furrows, and drainage channels using appropriate farm machinery.
- GS4.** be knowledgeable of different propagation methods and optimal timing for planting slips based on moisture content, precipitation, and soil conditions.
- GS5.** be skilled in installing and designing efficient irrigation or fertigation systems and ensuring proper water management through channel creation.
- GS6.** identify and address risks associated with lemongrass cultivation, including pest, disease, and weather-related concerns.
- GS7.** be proficient in operating tools and farm machinery such as ploughs, seeders, and irrigation systems for field preparation and planting.
- GS8.** be capable of working effectively with labor teams, guiding field staff, and coordinating with stakeholders to ensure timely and correct planting procedures.
- GS9.** maintain records of soil testing, land preparation activities, and planting schedules for continuous monitoring and improvement.
- GS10.** apply sustainable practices such as soil conservation, water-efficient irrigation, and environmentally safe land preparation methods.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. Interact with the Customer on their requirement and decide different types of plants for integrated farming	4	5	-	2
PC2. Ensure proper sunlight exposure for at least 6-8 hours a day.	4	5	-	4
PC3. Clear the land of debris and weeds.	4	5	-	2
PC4. Plough or till the soil to a depth of 8-10 inches.	3	5	-	4
PC5. Incorporate well-rotted organic matter to improve soil fertility.	3	4	-	2
PC6. Use healthy and disease free lemon grass saplings.	3	4	-	2
PC7. Plant at a spacing of 2-3 feet between rows and 1-2 feet between plants.	3	4	-	5
PC8. Plant during the rainy season or provide adequate irrigation.	3	4	-	5
PC9. Ensures proper soil conditioning and preparation for optimal plant growth	3	4	-	4
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0405
NOS Name	Plan and Suggest land Preparation and Planting
Sector	Agriculture
Sub-Sector	
Occupation	Field Crop
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/AGR/N0406: Suggest Nutrient and Irrigation Management

Description

This NOS trains learners to interpret reports for nutrient requirements and implement proper irrigation techniques. It emphasizes fertigation, organic nutrient use, and efficient water use to enhance crop productivity.

Scope

The scope covers the following :

- The scope of the NOS includes soil sample testing, identifying nutrient deficiencies, and recommending appropriate organic/inorganic fertilizers. It also involves planning and executing irrigation systems and scheduling, promoting sustainable practices like fertigation, mulching, and ensuring optimal soil health for targeted crop productivity.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** To Conduct soil tests before planting to determine nutrient levels.
- PC2.** To Adjust nutrient application based on the soil test results.
- PC3.** To Incorporate well-rotted organic matter into the soil during land preparation.
- PC4.** To Use compost or well decomposed manure to improve soil fertility.
- PC5.** To Apply nitrogen-rich organic fertilizers at the beginning of the growing season.
- PC6.** Use phosphorus-rich organic fertilizers during planting.
- PC7.** Consider applying bone meal or rock phosphate as a phosphorus supplement.
- PC8.** Include potassium-rich organic sources in the fertilization plan.
- PC9.** Ensure a balance of micronutrients by adding well balanced organic fertilizers.
- PC10.** Lemon grass requires consistent moisture throughout the growing season.
- PC11.** Adequate water is crucial during the establishment phase and flowering.
- PC12.** Consider installing drip irrigation systems for efficient water use.
- PC13.** Drip systems deliver water directly to the base of plants, minimizing wastage.
- PC14.** Apply mulch around lemon grass plants to reduce evaporation and retain soil moisture.
- PC15.** Mulching also helps control weeds, conserving water for the intended crops.
- PC16.** Develop a regular irrigation schedule, especially during dry periods.
- PC17.** Adjust the frequency based on local weather conditions and plant requirements.
- PC18.** Ensure proper drainage to prevent waterlogging, as lemon grass prefers well-drained soil.
- PC19.** Excess water can lead to root rot and other diseases.
- PC20.** Use clean and non-saline water for irrigation.
- PC21.** Saline water can adversely affect plant health and oil quality.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** the importance of testing soil samples through authorized laboratories to determine nutrient levels and soil health.
- KU2.** how to analyze lab reports to identify the micro and macronutrient needs of the soil and determine deficiencies.
- KU3.** how to select appropriate fertilizers (organic and inorganic) based on nutrient deficiencies and specific crop requirements.
- KU4.** fertigation techniques and the role of precision farming in applying nutrients effectively through irrigation systems.
- KU5.** about irrigation methods such as drip and basin irrigation, and how to manage water resources efficiently to enhance soil health.
- KU6.** how to comprehend the value of using organic matter, compost, and green manure to improve soil fertility and structure.
- KU7.** the principles of INM, combining chemical fertilizers, organic inputs, and bio-fertilizers for sustainable cultivation.
- KU8.** how to develop the ability to adjust nutrient plans based on re-testing of soil and changing crop needs during the growing season.
- KU9.** techniques for mulching to retain moisture, suppress weeds, and prevent soil erosion, thus supporting nutrient retention.
- KU10.** the safe handling and storage of fertilizers and chemicals, ensuring environmental compliance and personal safety during application.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Recommend nutrient application schedule based on crop stage.
- GS2.** interpret soil test reports to identify deficiencies and recommend suitable organic/inorganic nutrients and fertilizers.
- GS3.** select fertilizers based on soil needs and apply them in the right dosage and combinations to improve soil fertility and crop yield.
- GS4.** understand and implement efficient irrigation systems such as drip or basin irrigation, depending on the crop and terrain.
- GS5.** employ fertigation techniques to combine irrigation with nutrient delivery and use mulching to conserve soil moisture and suppress weed growth.
- GS6.** apply modern methods of precision farming to optimize resource use and enhance productivity.
- GS7.** recommend use of compost, green manure, and other organic inputs to boost soil health and sustainability.
- GS8.** collaborate with agronomists, lab technicians, and other professionals to validate analysis and ensure accuracy in nutrient management plans.
- GS9.** operate tools and systems like fertigation units, irrigation hoses, mulchers, and moisture meters effectively in the field.



Qualification Pack

- GS10.** assess crop performance and adjust irrigation and nutrient plans based on crop growth, climatic changes, and soil response.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. To Conduct soil tests before planting to determine nutrient levels.	2	3	-	-
PC2. To Adjust nutrient application based on the soil test results.	2	3	-	-
PC3. To Incorporate well-rotted organic matter into the soil during land preparation.	2	3	-	-
PC4. To Use compost or well decomposed manure to improve soil fertility.	2	3	-	-
PC5. To Apply nitrogen-rich organic fertilizers at the beginning of the growing season.	2	2	-	-
PC6. Use phosphorus-rich organic fertilizers during planting.	2	2	-	-
PC7. Consider applying bone meal or rock phosphate as a phosphorus supplement.	2	3	-	-
PC8. Include potassium-rich organic sources in the fertilization plan.	2	3	-	-
PC9. Ensure a balance of micronutrients by adding well balanced organic fertilizers.	2	3	-	-
PC10. Lemon grass requires consistent moisture throughout the growing season.	2	3	-	-
PC11. Adequate water is crucial during the establishment phase and flowering.	2	3	-	-
PC12. Consider installing drip irrigation systems for efficient water use.	2	3	-	-
PC13. Drip systems deliver water directly to the base of plants, minimizing wastage.	2	3	-	-
PC14. Apply mulch around lemon grass plants to reduce evaporation and retain soil moisture.	2	3	-	-
PC15. Mulching also helps control weeds, conserving water for the intended crops.	2	3	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Develop a regular irrigation schedule, especially during dry periods.	2	3	-	-
PC17. Adjust the frequency based on local weather conditions and plant requirements.	1	3	-	-
PC18. Ensure proper drainage to prevent waterlogging, as lemon grass prefers well-drained soil.	1	3	-	-
PC19. Excess water can lead to root rot and other diseases.	2	2	-	-
PC20. Use clean and non-saline water for irrigation.	2	3	-	-
PC21. Saline water can adversely affect plant health and oil quality.	2	3	-	-
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0406
NOS Name	Suggest Nutrient and Irrigation Management
Sector	Agriculture
Sub-Sector	
Occupation	Field Crop
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/AGR/N0407: Suggest Weed Control and Plant Protection

Description

This NOS involves identifying and managing weeds, pests, and diseases in lemongrass cultivation through safe and effective control methods. It focuses on minimizing losses and maintaining quality through integrated and sustainable plant protection practices.

Scope

The scope covers the following :

- Covers weed and pest monitoring, recommendation of control strategies, and supervision of safe application practices. It promotes integrated approaches to ensure healthy lemongrass production while protecting environmental and human health.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** To Apply a layer of organic mulch around lemon grass plants to suppress weed growth.
- PC2.** To and reduces competition from weeds, Mulching helps retain soil moisture
- PC3.** To Conduct regular hand weeding to remove weeds from the planting area.
- PC4.** To Ensure that weeds are removed before they reach maturity and produce seeds.
- PC5.** To Use mechanical cultivation methods such as shallow tilling to control weeds.
- PC6.** Be cautious not to damage the lemon grass roots during cultivation.
- PC7.** Consider using preemergence herbicides before planting to control weed germination.
- PC8.** Use herbicides selectively and follow recommended application rates.
- PC9.** Encourage the presence of natural predators, such as beneficial insects, to control pest populations.
- PC10.** Avoid the indiscriminate use of pesticides that may harm beneficial organisms.
- PC11.** Use neem oil as a natural and organic pesticide.
- PC12.** Neem oil has insecticidal properties and can help control pests.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** various types of weeds affecting crops, their growth cycles, and the competitive impact they have on crop productivity.
- KU2.** knowledge of manual, mechanical, and chemical weed control techniques, including the appropriate use of tools like weed trimmers and safe application of herbicides.
- KU3.** identification of common pests and diseases affecting crops like lemongrass or dragon fruit, along with visible symptoms, pest life cycles, and seasonal occurrence (pest/disease calendars).

Qualification Pack

- KU4.** the use of biological controls, cultural practices, and judicious pesticide use to minimize ecological impact.
- KU5.** about crop hygiene, pruning techniques, and use of disease-resistant varieties as preventive and curative measures.
- KU6.** the knowledge of safe handling, dosage, timing, and storage of pesticides, fungicides, and herbicides, adhering to manufacturer's instructions and environmental guidelines.
- KU7.** the awareness of biological control agents like ladybugs, parasitoid wasps, and microbes for natural pest suppression.
- KU8.** the importance of disinfecting equipment such as pruning tools to prevent the spread of infection between plants.
- KU9.** nutrient-related disorders in plants and suggest corrective actions like soil amendments (e.g., gypsum or sulfur) based on lab results.
- KU10.** the knowledge of removal and safe disposal of infected plant material and maintenance of field cleanliness to reduce disease spread.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** recognize different types of weeds, pests, and diseases affecting lemongrass and other crops through field observation and knowledge of their lifecycle and symptoms.
- GS2.** diagnose crop diseases, understand physiological disorders, and implement preventive and curative measures using resistant varieties, pruning, and sanitation practices.
- GS3.** be skilled in implementing integrated pest and disease management strategies by combining cultural, mechanical, biological, and chemical methods to minimize ecological impact.
- GS4.** understand the importance of maintaining environmental balance by promoting use of organic mulching, compost, and beneficial insects in pest management practices.
- GS5.** be competent in maintaining records of pesticide usage, disease occurrence, weed control schedules, and treatment effectiveness for traceability and improvement.
- GS6.** operate and maintain equipment such as weed trimmers, sprayers, pruning tools, and soil testing kits effectively and safely.
- GS7.** demonstrates good judgment in deciding the appropriate time and method for applying plant protection measures based on climatic conditions, crop stage, and pest threat levels.
- GS8.** be able to train farm workers on weed and pest identification and communicate clearly about plant protection measures to be undertaken for sustainable crop health.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. To Apply a layer of organic mulch around lemon grass plants to suppress weed growth.	4	5	-	-
PC2. To and reduces competition from weeds, Mulching helps retain soil moisture	3	5	-	-
PC3. To Conduct regular hand weeding to remove weeds from the planting area.	3	5	-	-
PC4. To Ensure that weeds are removed before they reach maturity and produce seeds.	3	5	-	-
PC5. To Use mechanical cultivation methods such as shallow tilling to control weeds.	4	5	-	-
PC6. Be cautious not to damage the lemon grass roots during cultivation.	3	5	-	-
PC7. Consider using preemergence herbicides before planting to control weed germination.	3	5	-	-
PC8. Use herbicides selectively and follow recommended application rates.	4	5	-	-
PC9. Encourage the presence of natural predators, such as beneficial insects, to control pest populations.	3	5	-	-
PC10. Avoid the indiscriminate use of pesticides that may harm beneficial organisms.	4	5	-	-
PC11. Use neem oil as a natural and organic pesticide.	3	5	-	-
PC12. Neem oil has insecticidal properties and can help control pests.	3	5	-	-
NOS Total	40	60	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0407
NOS Name	Suggest Weed Control and Plant Protection
Sector	Agriculture
Sub-Sector	
Occupation	Field Crop
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/AGR/N0408: Post - Harvest storage and Lemongrass oil extraction

Description

This NOS focuses on the supervision of post-harvest handling, storage, and oil extraction of lemongrass. It includes planning for timely harvest, quality maintenance, and the safe and efficient extraction of lemongrass oil for commercial use.

Scope

The scope covers the following :

- Involves post-harvest activities such as bundling, transporting, distilling, filtering, storing, and packaging lemongrass oil. The scope includes ensuring quality, minimizing losses, and improving the value of lemongrass through efficient processing and handling.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** To Harvest lemon grass when it reaches a height of about 2 to 3 feet.
- PC2.** Use sharp and clean pruning shears or a sickle for harvesting.
- PC3.** Cut the leaves above the ground level, leaving the roots intact for regrowth.
- PC4.** To Perform selective harvesting by cutting mature leaves while leaving young and healthy leaves for continued growth.
- PC5.** Harvest in the morning when the oil content in the leaves is higher.
- PC6.** Harvest every 3 to 4 months, depending on the growth rate and climate.
- PC7.** Adjust the frequency based on the health and growth of the lemon grass.
- PC8.** Dry the harvested lemon grass in a well-ventilated area away from direct sunlight.
- PC9.** to preserve the essential oils, Use a drying rack or hang bundles upside down
- PC10.** To retain freshness and aroma. Store dried lemon grass leaves in airtight containers
- PC11.** To avoid contamination, Glass or metal containers are preferred
- PC12.** Store the containers in a cool, dark place to preserve the quality of the dried leaves.
- PC13.** To prevent mold, Protect the lemongrass from moisture
- PC14.** Lemongrass oil is commonly extracted through steam distillation.
- PC15.** To separate the essential oil from the plant material, Use a distillation apparatus
- PC16.** Place the lemongrass in the distillation apparatus, and steam is passed through.
- PC17.** The essential oil is captured, and water condenses separately.
- PC18.** Allow the collected oil to settle, and separate it from the water layer.
- PC19.** The resulting oil is pure lemongrass essential oil.
- PC20.** Store the extracted lemongrass oil in dark-colored glass bottles.
- PC21.** Seal the bottles tightly to prevent exposure to air and light.

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** the ideal conditions and procedures for post-harvest handling of lemongrass to retain oil content and quality.
- KU2.** the parameters for short and long-term storage including moisture levels, temperature control, and protection from pests or microbial growth.
- KU3.** theoretical understanding of steam distillation process, the most common method used for extracting lemongrass essential oil.
- KU4.** the familiarity with distillation units, including condensers, separators, and boilers used in essential oil extraction.
- KU5.** the quality indicators of lemongrass oil such as color, aroma, density, and chemical composition (e.g., citral content).
- KU6.** the methods for chopping, drying, and feeding lemongrass into the distillation equipment for efficient oil yield.
- KU7.** how to handle biomass residue after oil extraction in an eco-friendly manner, either for composting or bioenergy use.
- KU8.** the techniques for filtering, storing, and packaging essential oil using dark glass bottles and airtight containers to prevent degradation.
- KU9.** occupational health and safety norms during distillation to avoid risks like burns, chemical exposure, or machinery hazards.
- KU10.** the basics of labeling, batch coding, and maintaining records of oil extracted for traceability and quality assurance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** assess the maturity of lemongrass based on standard indicators like leaf size, aroma, and growth duration before initiating harvesting.
- GS2.** handle lemongrass post-harvest to avoid contamination, ensure moisture control, and prepare it properly for oil extraction.
- GS3.** be proficient in operating steam distillation units, including monitoring temperature, pressure, and time to maximize oil yield and maintain quality.
- GS4.** store lemongrass oil in appropriate containers (preferably stainless steel or aluminum) to prevent degradation due to light, air, or temperature.
- GS5.** testing the extracted oil for purity, aroma profile, and other quality parameters using both sensory and basic lab techniques.
- GS6.** the practical skills in cleaning, maintaining, and troubleshooting extraction and storage equipment to ensure safety and operational efficiency.
- GS7.** do hygienic packaging standards and proper labelling practices, including batch numbers, production dates, and quality grades.
- GS8.** maintain detailed records of raw material input, oil yield, processing time, and storage logs to ensure traceability and compliance.
- GS9.** handle and dispose, residual biomass post-extraction in an eco-friendly manner, possibly for compost or fuel.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. To Harvest lemon grass when it reaches a height of about 2 to 3 feet.	-	5	-	-
PC2. Use sharp and clean pruning shears or a sickle for harvesting.	2	5	-	-
PC3. Cut the leaves above the ground level, leaving the roots intact for regrowth.	2	5	-	-
PC4. To Perform selective harvesting by cutting mature leaves while leaving young and healthy leaves for continued growth.	2	5	-	-
PC5. Harvest in the morning when the oil content in the leaves is higher.	-	5	-	-
PC6. Harvest every 3 to 4 months, depending on the growth rate and climate.	-	5	-	-
PC7. Adjust the frequency based on the health and growth of the lemon grass.	2	4	-	-
PC8. Dry the harvested lemon grass in a well-ventilated area away from direct sunlight.	2	4	-	-
PC9. to preserve the essential oils, Use a drying rack or hang bundles upside down	2	4	-	-
PC10. To retain freshness and aroma. Store dried lemon grass leaves in airtight containers	2	4	-	-
PC11. To avoid contamination, Glass or metal containers are preferred	1	2	-	-
PC12. Store the containers in a cool, dark place to preserve the quality of the dried leaves.	3	2	-	-
PC13. To prevent mold, Protect the lemongrass from moisture	-	2	-	-
PC14. Lemongrass oil is commonly extracted through steam distillation.	2	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. To separate the essential oil from the plant material, Use a distillation apparatus	2	2	-	-
PC16. Place the lemongrass in the distillation apparatus, and steam is passed through.	2	2	-	-
PC17. The essential oil is captured, and water condenses separately.	2	2	-	-
PC18. Allow the collected oil to settle, and separate it from the water layer.	1	3	-	-
PC19. The resulting oil is pure lemongrass essential oil.	1	3	-	-
PC20. Store the extracted lemongrass oil in dark-colored glass bottles.	1	3	-	-
PC21. Seal the bottles tightly to prevent exposure to air and light.	1	1	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0408
NOS Name	Post - Harvest storage and Lemongrass oil extraction
Sector	Agriculture
Sub-Sector	
Occupation	Field Crop
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	30/01/2027
NSQC Clearance Date	31/01/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

Qualification Pack

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
 - Questions are mapped to the specified assessment criteria.
 - Assessor must be ToA certified & trainer must be ToT Certified
4. Types of evidence or evidence-gathering protocol:
- Time-stamped & geotagged reporting of the assessor from assessment location.
 - Centre photographs with signboards and scheme specific branding
 - Video Call verification/online proctoring will be done during the assessment from AB.
5. Method of verification or validation:
- Surprise visit to the assessment location
6. Minimum Percentage Required
- 70% overall to pass.
 - Minimum 70% in each NOS
7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.
8. Assessor Details: ToA Certified Assessor with relevant experience.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N0409.Implement Health and Safety standards at Work Place	30	40	-	30	100	10

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	50	70	-	30	150	20

Elective: 1 Dragon Fruit Supervisor

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N0401.Plan and Deploy planting materials	30	40	-	30	100	20
CUTM/AGR/N0402.Suggest optimal cultivation practices with integrated farming	40	60	-	-	100	20
CUTM/AGR/N0403.Suggest integrated nutrient management in Dragon fruit cultivation	40	60	-	-	100	20
CUTM/AGR/N0404.Suggest integrated disease and pest management in Dragon fruit cultivation	30	70	-	-	100	20
Total	140	230	-	30	400	80

Elective: 2 Lemon Grass Supervisor

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N0405.Plan and Suggest land Preparation and Planting	30	40	-	30	100	20
CUTM/AGR/N0406.Suggest Nutrient and Irrigation Management	40	60	-	-	100	20



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N0407.Suggest Weed Control and Plant Protection	40	60	-	-	100	20
CUTM/AGR/N0408.Post - Harvest storage and Lemongrass oil extraction	30	70	-	-	100	20
Total	140	230	-	30	400	80



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.